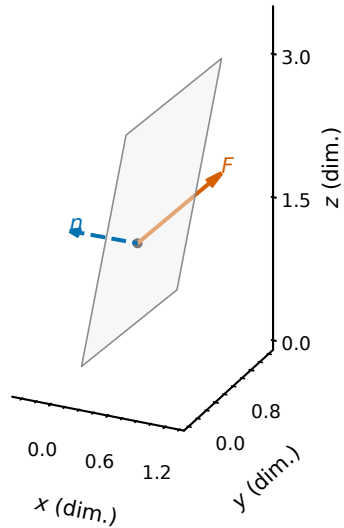
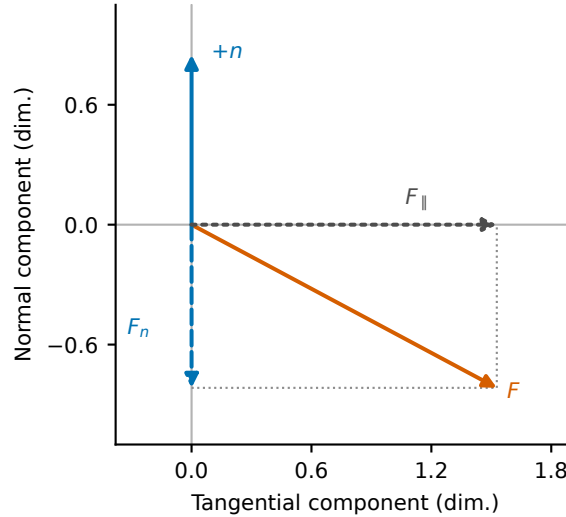
A



$n = (-1, -2, 1)/\sqrt{6}$; $F = (1, 1, 1)$

B



$F \cdot n = -2/\sqrt{6}$; $d\Phi = -2\,du\,dv$